Finite-size decay-mode hybridization near a mean-field fold in an open Rydberg ensemble

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Predicting relaxation near a mean-field fold requires connecting decay-mode mixing to physical preparation and detection. We establish this connection in a uniformly interacting Rydberg ensemble with independent local decay, retaining the complete permutation-invariant operator space. Two real modes exchange high-branch relaxation residues at an avoided crossing on the monostable side of the fold. Their separation decreases approximately exponentially over $N = 24$–56, while their common decay rate remains finite. Exact coherence elimination recovers both eigenvalues as self-consistent roots of one effective branch. The initial-coherence source and spectral-derivative normalization then recover their observable residues. Fixed-preparation scans at $N = 24$–48 resolve the distinction between mode exchange and density response. Over $0 \leq \gamma t \leq 12$, pronounced residue redistribution accompanies smooth complete density signals. The coherence-to-population peak stays below the independent-atom reference, while its full time dependence differs from mean field. These results connect finite-size spectral hybridization, coherence memory, and preparation-dependent relaxation within one many-body description.

I. INTRODUCTION

Driven Rydberg ensembles connect nonlinear mean-field bistability with finite quantum dynamics [1–4]. A stable mean-field branch terminates at a saddle-node fold, whereas a finite quantum system evolves under a linear, trace-preserving Liouvillian. Connecting the two descriptions requires both the many-body decay spectrum and the response selected by physical preparation and detection.

Exceptional-point (EP) interpretations of nonlinear switching motivate separating stationary-root singularities from dynamical eigenvalue coalescence [4–6]. A fold of stationary solutions need not be a defective eigenvalue of the mean-field Jacobian. Bounded finite-dimensional quantum evolution also excludes nontrivial Jordan blocks in the stationary sector [7]. Metastability can instead arise from low-lying decay modes separated from faster relaxation [8, 9], or from large eigenmode-expansion coefficients that prolong transients [10, 11]. Observable relaxation therefore depends on spectral weights as well as decay rates.

For uniformly interacting spins with local decay, collective switching and steady-state correlations are established [1, 2], and an exact finite-N steady state is available [12]. Exponentially slow switching is central to first-order dissipative transitions [13]; quantum-instanton methods predict switching rates for collective spin systems, including local dissipation [14, 15]. A related conjecture connects a symmetry-selected steady-state purification potential to the ordinary Liouvillian gap [16]. These approaches address relaxation toward the stationary sector. The response associated with mixing among modes at a finite common decay rate calls for a separate spectral analysis.

Finite-size avoided crossings from mode mixing [17] and few-mode descriptions of hysteresis [18] provide a starting point. Cabot, Giorgi, and Zambrini connected a Liouvillian EP to a mean-field bifurcation between phase diffusion and quantum activation in a driven oscillator [19]. Collective-spin models exhibit spectral doublets approaching finite nonzero decay rates [20], with preparation-dependent visibility and cancelling spectral residues [21]. For the locally damped Rydberg ensemble, the response problem is how a real-mode avoided crossing redistributes observable weights and shapes the complete density transient.

At $\Omega = 2\gamma$ and $V = 10\gamma$, we resolve this connection with all permutation-invariant spin blocks and local-decay transfers retained. Physical preparations reveal an exchange of signed relaxation residues between the two real modes. Biorthogonal projection identifies their invariant coupling product; exact coherence elimination connects their roots to response weights through the initial-coherence source. Fixed-preparation scans show that this spectral redistribution coexists with smooth density transients, whose time dependence retains many-body effects beyond mean field.

II. MODEL, MEAN-FIELD FOLD, AND FINITE-SIZE SPECTRUM

We consider N two-level atoms with excited-state number $\hat{N}_e = \sum_i \hat{n}_i$, collective spin $\hat{J}_\alpha = \frac{1}{2}\sum_i \hat{\sigma}_i^\alpha$, and independent decay:

$$\dot{\rho} = \mathcal{L}_N \rho = -i[\hat{H}, \rho] + \gamma \sum_i \mathcal{D}[\hat{\sigma}_i^-]\rho, \tag{1}$$

$$\hat{H} = -\Delta \hat{N}_e + \Omega \hat{J}_x + \frac{V}{2N}\hat{N}_e(\hat{N}_e - 1), \tag{2}$$

where $\mathcal{D}[c]\rho = c\rho c^\dagger - \{c^\dagger c, \rho\}/2$. This is the uniformly interacting, locally damped model of Refs. [1, 2], within the exactly solvable steady-state family of Ref. [12]. The Supplemental Material (SM) gives the finite-N parameter mappings, including the detuning shift associated with $\hat{N}_e(\hat{N}_e - 1)$. We set $\gamma = 1$, so frequencies and eigenvalues below are measured in units of γ, and time in γ^{-1}.

Permutation invariance reduces the operator space to dimension $\binom{N+3}{3}$ [22]. We retain every Dicke j block and the transitions between blocks caused by local decay. This differs from the maximum-spin effective-jump approximation used in recent Rydberg switching calculations [23]. Throughout, "full generator" denotes this complete permutation-invariant operator space, which contains the product states and observables considered here. It does not enumerate modes in other permutation-symmetry sectors.

In the large-N factorized limit [1, 2], write the single-atom coherence as $\rho_{eg} = u+iv$ and the excited population as n. The mean-field equations are

$$\begin{aligned} \dot{n} &= -n - \Omega v, \\ \dot{u} &= -(\Delta - Vn)v - u/2, \\ \dot{v} &= (\Delta - Vn)u - \Omega(1-2n)/2 - v/2. \end{aligned} \tag{3}$$

Their stationary populations satisfy

$$F(n) = n[(\Delta - Vn)^2 + 1/4 + \Omega^2/2] - \Omega^2/4 = 0. \tag{4}$$

At $\Omega = 2$, $V = 10$, the left fold is at $(\Delta_f, n_f) \simeq (3.60, 0.157)$. Using $F = \partial_n F = 0$, the characteristic polynomial of the Bloch Jacobian at a fold reduces to

$$\chi(\lambda) = \lambda[(\lambda+1)^2 + \Omega^2/2]. \tag{5}$$

Thus the zero eigenvalue is simple, including when the fold reaches the cusp. Li, Sun, and Yi [6] obtained square-root and cube-root singularities of stationary excitation densities near fold and cusp points. Those exponents describe roots of the nonlinear stationary equation; Eq. (5) concerns the distinct linear stability problem. In particular, a triple stationary root does not imply an EP3 of this Jacobian. The coalescing fixed points and the surviving high-population branch must also be distinguished; Eq. (5) describes the former.

We locate a local minimum of the separation between two real decay eigenvalues and denote it by $\Delta_*(N)$. Define

$$g_N = \frac{\lambda_+(\Delta_*) - \lambda_-(\Delta_*)}{2} > 0. \tag{6}$$

Across $N = 24, 28, \ldots, 56$, the crossing drifts toward the fold while remaining on its monostable side, and the half-gap decreases by almost two orders of magnitude [Fig. 1]. An exponential fit gives $\sigma \simeq 0.14$, with a small systematic drift as the size window changes (SM). We use this slope to characterize the observed window; the limiting exponent and convergence of $\Delta_*(N)$ to Δ_f remain undetermined.

The half-gap resolves hybridization within the decaying spectrum. At the $N = 32$ crossing, $g_N \simeq 0.01$ while the mean decay rate is $-\bar{\lambda} \simeq 0.75$, where $\bar{\lambda} = (\lambda_+ + \lambda_-)/2$. An oscillatory pair decays more slowly than both real modes (SM). Thus g_N measures excited-mode separation, whereas the ordinary Liouvillian gap $\Gamma_{\mathrm{diss}} = -\max_{\lambda\neq 0} \mathrm{Re}\,\lambda$ measures decay toward the stationary sector.

III. PHYSICAL RESPONSE AND MODE CHARACTER

A Hermitian right eigenoperator at a nonzero decay eigenvalue is traceless; its signed excitation-number profile is not a probability distribution. We test mode character through observable residues, whose dependence on both preparation and detection is essential in nonnormal dynamics [10, 21].

Let R_i and W_i be right and transpose-left eigenvectors of the stored generator, normalized by $W_i^T R_j = \delta_{ij}$. The transpose denotes the algebraic dual of the chosen vectorization. For the excitation-density observable $o^T\rho = \langle \hat{N}_e \rangle / N$, an initial difference $\delta\rho$ gives the exact contribution of a simple mode,

$$\delta n_i(t) = a_i e^{\lambda_i t}, \qquad a_i = (o^T R_i)(W_i^T \delta\rho). \tag{7}$$

Each residue is invariant under reciprocal rescaling of its left and right eigenvectors. These signed weights are not probabilities of physical states or metastable phases.

We use two physical preparations, whose construction is specified in the SM. The first subtracts positive product states displaced slightly in opposite directions along the real relaxation direction of the high mean-field fixed point. This high-branch preparation is reconstructed at each detuning to follow local mode character. The second subtracts the exact stationary state from a fixed product state prepared at the left-fold Bloch vector. It probes low-population response without assuming a finite-size metastable state outside the bistable interval.

Across the crossing, the dominant high-branch response transfers from λ_+ to λ_- [Fig. 2(a)]. The exchange persists in a separate $N = 40$ calculation, and reducing the preparation displacement confirms the perturbative response (SM).

The fold-state preparation excites both modes, with large opposite residues near the crossing [Fig. 2(b)]. Their cancellation preserves a bounded physical signal, as analyzed for collective-spin exceptional spectral phases [21]. Here the cancellation is resolved at finite positive half-gap; the SM specifies the additional limits needed for a $1/g_N$ residue asymptote or a real-axis Jordan form.

The signed excitation-number profile $P_i(k) = \mathrm{Tr}(\Pi_k R_i)$ provides a complementary diagnostic. Its absolute-profile fraction below a fixed low-density cutoff exchanges between the branches, alongside the high-branch response [Fig. 2(c)]. Both profiles retain weight outside this region. The two diagnostics jointly identify an exchange of high-branch response and low-density character between extended decay modes.

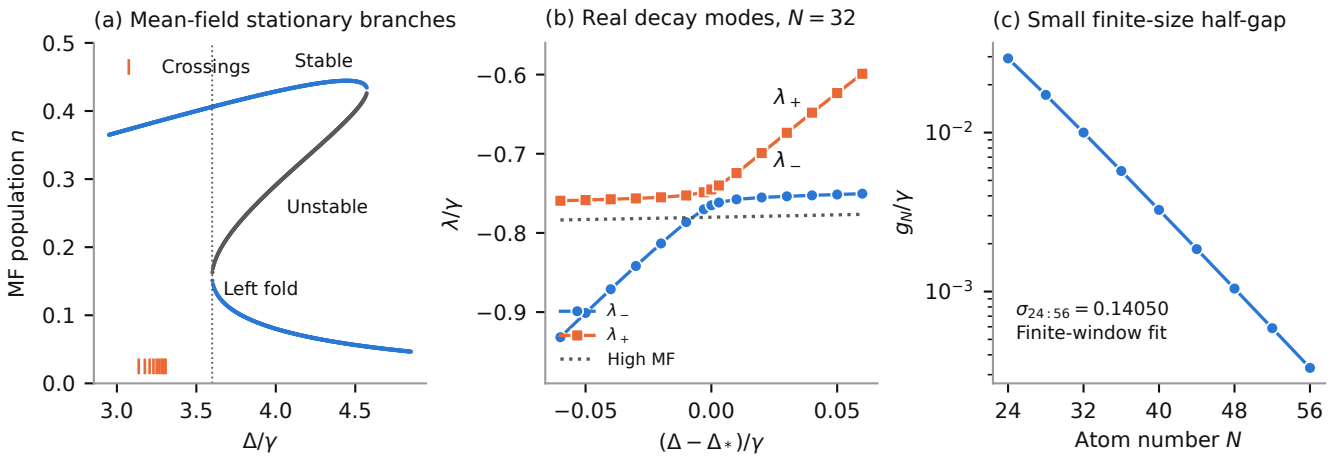


FIG. 1. Mean-field geometry and the finite-size real-mode crossing at $\Omega = 2$, $V = 10$. (a) Stationary populations: stable branches in blue, unstable branch in gray. Orange ticks mark the real-axis crossing positions for $N = 24, 28, \ldots, 56$; all lie on the monostable side of the left fold. (b) The two target real eigenvalues for $N = 32$, ordered as $\lambda_- < \lambda_+$, versus $\delta = \Delta - \Delta_*$. The dotted curve is the real relaxation eigenvalue of the surviving high mean-field branch. The modes exchange high-branch response, but the finite-size rates need not equal the mean-field rate. (c) Half-gap at its local minimum. The quoted exponent is a finite-window fit, not an established large-N limit.

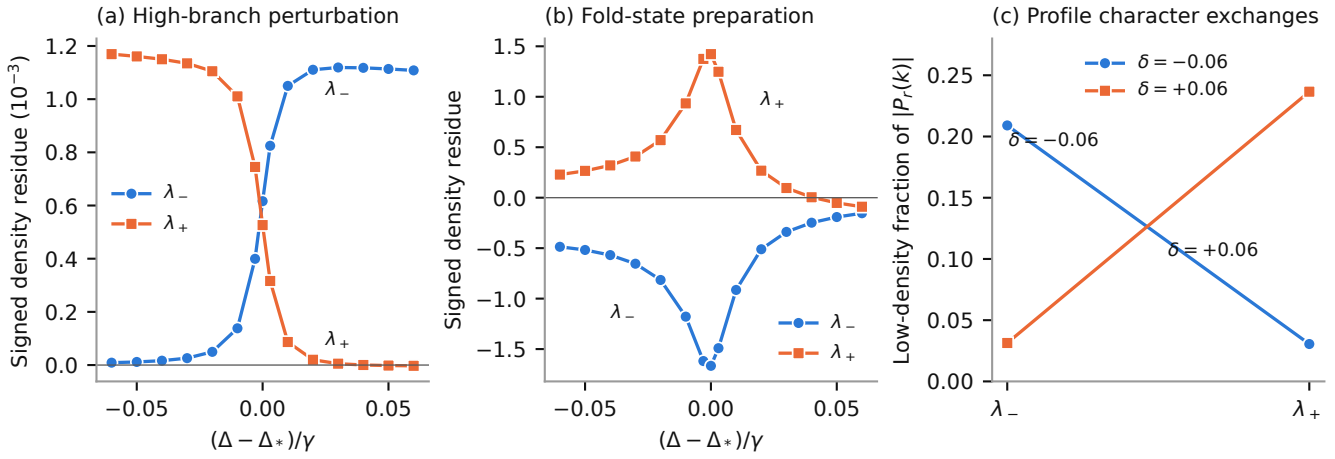


FIG. 2. Physical residues and exchange of mode character, $N = 32$. (a) Residues for product-state differences about the high mean-field fixed point, reconstructed at each detuning. (b) Residues for a fixed fold product state relative to the current stationary state. Large opposite residues cancel in the physical signal. (c) Fraction of the absolute excitation-number profile below $k/N = 0.22$ at the two endpoints. This normalization-independent ratio characterizes a signed eigenoperator, not a population probability of a physical state.

IV. BIORTHOGONAL TWO-MODE REDUCTION AND COMPLEX-DETUNING EPS

A. Local reduction at the gap minimum

Write $\mathcal{L}_N(\Delta) = L_0 + \Delta L_1$, with $L_1\rho = i[\hat{N}_e, \rho]$. At Δ_*, choose bases R, W of the exact two-dimensional right and left invariant subspaces with $W^T R = I$. Diagonalizing $W^T L_1 R$ defines a local diabatic basis with distinct real slopes s_a, s_b. Holding that basis fixed gives

$$H_P(\delta) = \begin{pmatrix} e_a + s_a\delta & V_{ab} \\ V_{ba} & e_b + s_b\delta \end{pmatrix}, \qquad \delta = \Delta - \Delta_*. \quad (8)$$

Here a denotes the smaller-absolute-slope branch. At $\delta = 0$ the projection is exact. Coupling to the complementary invariant subspace produces a spectral self-energy starting at order δ^2 (SM); its small absolute size does not alone establish its smallness relative to an exponentially small coupling.

The usual two-by-two discriminant [24] gives the

squared separation and its derivative at the projection point:

$$d^2 = (e_a - e_b)^2 + 4V_{ab}V_{ba}, \qquad \partial_\Delta d^2 = 2(e_a - e_b)(s_a - s_b). \tag{9}$$

At a nondegenerate minimum of a positive real gap, distinct slopes imply $e_a = e_b$ and hence

$$\boxed{g_N^2 = V_{ab}V_{ba}.} \tag{10}$$

The product is a normalization-independent measure of hybridization within the exact finite-size spectral subspace. Predicting its size dependence independently requires an asymptotic calculation.

Under $R_i \mapsto c_i R_i$, $W_i \mapsto c_i^{-1} W_i$,

$$V_{ab} \mapsto (c_b/c_a)V_{ab}, \qquad V_{ba} \mapsto (c_a/c_b)V_{ba}. \tag{11}$$

The product isolates the physical splitting from the normalization of individual modes. If c_i depend on N, individual coupling exponents change with that convention. Switching from population-peak to physical trace-norm normalization changes both exponents while preserving their sum, 2σ. A balanced convention makes each equal to σ by construction [Fig. 3; definitions and values in the SM]. Accordingly, the invariant object for an activation interpretation is the product, while each directional exponent retains its normalization dependence.

The weak matrix element contains cancellations involving coherences: at $N = 40$, its output-diagonal and output-off-diagonal contributions cancel by almost four orders of magnitude (SM). This partition does not separate two invariant physical processes. Since $W_b^T(L - cI)R_a = W_b^T L R_a$ for any scalar c, choosing $c = e_a$ transfers the cancellation into the small residual $(L - e_a)R_a$. A right population tail alone therefore does not determine the full biorthogonal matrix element; the boundary-flux analysis in the SM reaches the same limitation.

B. Analytic continuation in detuning

We separately continue the full generator as $L_0 + \zeta L_1$ for complex detuning ζ. This continuation uses $-i[H(\zeta), \rho]$; it is not $-i[H\rho - \rho H^\dagger]$. The continued evolution preserves trace but generally not Hermiticity. It is therefore not a completely positive and trace-preserving (CPTP) physical map.

Full-generator continuation locates nearby complex EPs for $N = 24, 32, 40$. At each point, the projected traceless matrix is nonzero and squares to zero within numerical precision. At $N = 32$, encircling the EP exchanges the two eigenvalues after one circuit and restores them after two (coordinates and stability checks in the SM).

The local linear model predicts

$$|\operatorname{Im}\zeta_{\mathrm{EP}}| \simeq \frac{2g_N}{|s_a - s_b|}, \tag{12}$$

in subpercent agreement with the continued spectra at all three sizes. The complex EP distance thus tracks the real-axis mixing scale for these crossings. This local result does not imply a nearby EP for every avoided crossing or a physical degeneracy at real parameters.

V. EXACT ELIMINATION OF COHERENCE SECTORS

A. Spectral-parameter-dependent effective operator

We use projection methods that retain the spectral parameter [25, 26], evaluated by matrix continued fractions as in spin master equations [27]. Group the Dicke operator basis by signed coherence order $q = m - m'$. The diagonal Hamiltonian and local decay preserve q, whereas the drive connects q only to $q \pm 1$. This gives a block-tridiagonal generator while retaining every Dicke j block.

Let $A_q = L_{q,q}$, $U_q = L_{q,q+1}$, and $D_q = L_{q+1,q}$. Starting at the highest positive order $q_{\mathrm{top}} = N$, define

$$\begin{aligned} &\Sigma_{q_{\mathrm{top}}}(z) = 0, \\ &\Sigma_{q-1}(z) = U_{q-1}[zI - A_q - \Sigma_q(z)]^{-1} D_{q-1}. \end{aligned} \tag{13}$$

Here z is an eigenvalue variable, distinct from the complex detuning ζ above. For real physical parameters and real z, Hermitian conjugation relates the positive and negative coherence chains. Their sum yields

$$K_N(z) = A_0 + 2\operatorname{Re}\Sigma_0(z), \qquad K_N(z)r_0 = z r_0. \tag{14}$$

The $q = 0$ sector still contains the populations of every Dicke block; it is not the excitation-number marginal alone, and $K_N(z)$ need not be a classical Markov generator.

With P projecting onto $q = 0$ and $Q = I - P$, the Schur identity reads

$$\det(zI - L) = \det(zI - L_{QQ}) \det[zI - K_N(z)]. \tag{15}$$

It applies where $z - L_{QQ}$ is invertible; the recursion also requires invertible intermediate denominators. Purely coherent modes with zero $q = 0$ projection and points at elimination poles require separate treatment. For complex z, the analytic continuation involves $\Sigma_+(z) + \overline{\Sigma_+(\bar{z})}$, not $2\operatorname{Re}\Sigma_+(z)$. The complex-detuning EPs above are computed separately from the full generator.

The spectral dependence has a time-domain counterpart in the projection-operator memory formalism [28, 29]. Direct elimination of x_Q from our linear equations gives

$$\begin{aligned} \dot{x}_0(t) = L_{00}x_0(t) + L_{0Q}e^{tL_{QQ}}x_Q(0) \\ + \int_0^t L_{0Q}e^{(t-s)L_{QQ}}L_{Q0}x_0(s)\,ds. \end{aligned} \tag{16}$$

Thus the self-energy represents memory carried by the eliminated coherence sectors. Replacing it by a constant is an additional approximation.

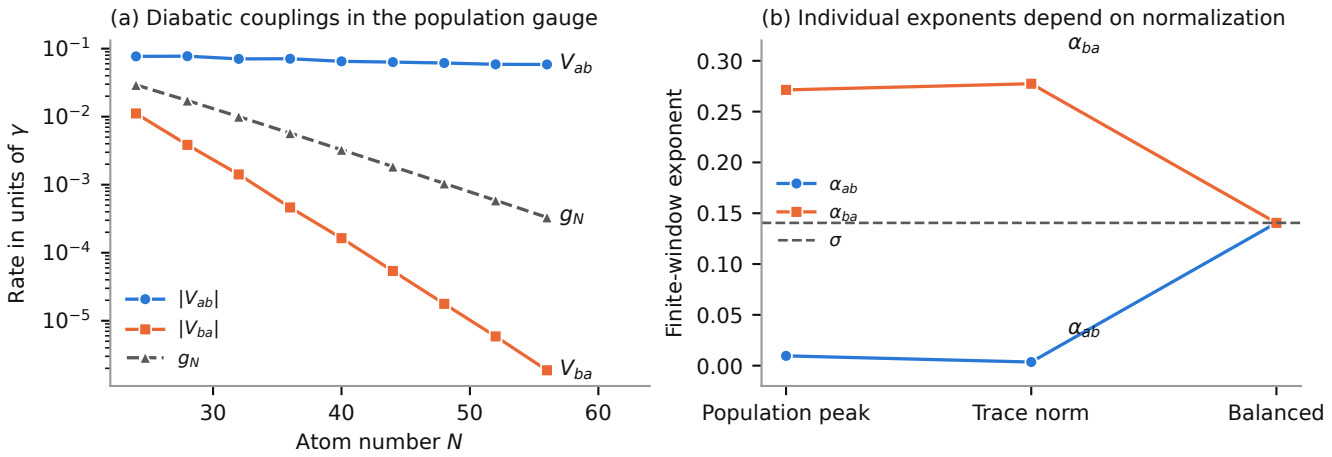


FIG. 3. Biorthogonal gap decomposition. (a) Couplings in the population-peak convention, together with the invariant half-gap. (b) Finite-window exponents under three normalizations of the same two-dimensional subspace. The physical trace-norm convention changes the individual exponents; balancing the couplings makes each equal to σ by construction. The invariant statement is their product and the sum of their logarithmic slopes.

B. Two roots of one effective branch

In the local real spectral window, select a simple real eigenvalue branch $\mu_N(z)$ of $K_N(z)$ and define $f_N(z) = \mu_N(z) - z$. We find a negative local minimum,

$$f_N(z_c) = -\epsilon_N < 0, \qquad f'_N(z_c) = 0, \\ c_N = f''_N(z_c) > 0. \tag{17}$$

Its two nearby roots give the two decay eigenvalues. Expanding around the minimum gives

$$f_N(z) = -\epsilon_N + \frac{c_N}{2}(z - z_c)^2 + \cdots, \qquad g_{N,\mathrm{quad}} = \sqrt{\frac{2\epsilon_N}{c_N}}. \tag{18}$$

The derivatives are calculated analytically through the matrix recursion and the non-Hermitian effective eigenproblem (SM).

For $N = 24, 32, 40, 48$, direct roots of $f_N = 0$ reproduce the independent full-generator pair; back-substitution and numerical checks are given in the SM. The quadratic approximation has percent-level accuracy at the smallest size and improves rapidly with N [Fig. 4]. Higher-order terms also shift the root midpoint away from z_c.

Self-consistency in z produces the two nearby roots of one effective branch. At $N = 48$, freezing K_N at z_c leaves only one eigenvalue near the full-generator pair (SM). Spectral dependence therefore carries the splitting that this frozen operator cannot resolve.

Over the computed window, the minimum depth falls rapidly while the curvature changes modestly [Fig. 4]. Equation (18) connects this depth-to-curvature ratio, ϵ_N/c_N, directly to the finite-size half-gap. Both factors are evaluated from the finite-size spectrum; an independent large-N prediction requires their asymptotic scaling, including the curvature prefactor.

C. Connection to biorthogonal self-orthogonality

The effective spectral slope has a precise relation to the full-mode norm. Write $G = (zI - L_{QQ})^{-1}$, $B = L_{0Q}$, and $C = L_{Q0}$, so that $K = L_{00} + BGC$ and $K_z = -BG^2C$. For a simple effective eigenbranch with $Kr = \mu r$, $l^T K = \mu l^T$, and $l^T r = 1$, define the back-substituted vectors

$$R = \begin{pmatrix} r \\ GCr \end{pmatrix}, \qquad W^T = (l^T,\, l^T BG). \tag{19}$$

Then

$$\boxed{W^T R = l^T(I - K_z)r = 1 - \mu_z = -f_z.} \tag{20}$$

At a root, this is the standard derivative pairing for the nonlinear eigenvalue problem $T(z) = zI - K(z)$ [26, 30]. At the negative minimum, $f_z = 0$ but $f \neq 0$: the back-substituted vectors are not eigenmodes, so this is not an EP. Under regular elimination and a simple effective branch, $f = f_z = 0$ with $f_{zz} \neq 0$ would instead give an EP2 (SM). This criterion does not determine the observed depth ϵ_N.

D. Initial coherences and observable residues

The same elimination must retain the initial-coherence source in Eq. (16). For an initial operator $x = (x_0, x_Q)^T$, its reduced Laplace transform obeys

$$[zI - K(z)]\widetilde{x}_0(z) = b(z), \qquad b(z) = x_0 + BG(z)x_Q. \tag{21}$$

For the population observable, $o^T = (o_0^T, 0)$, the standard simple-pole residue [26, 30] becomes

$$a_i = \frac{(o_0^T r_i)[l_i^T b(\lambda_i)]}{l_i^T[I - K'(\lambda_i)]r_i}. \tag{22}$$

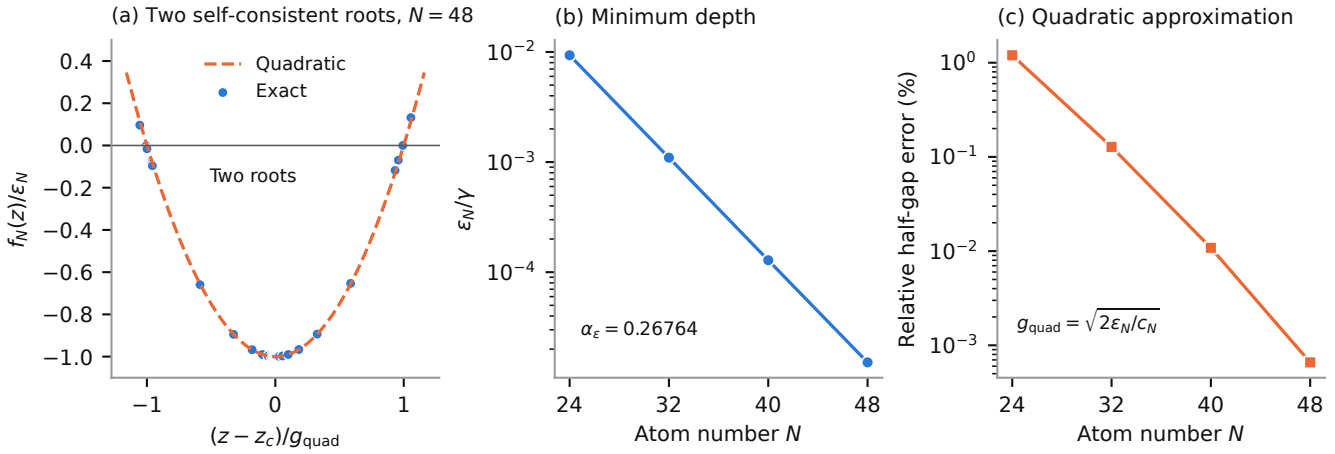


FIG. 4. Exact coherence elimination at the supplied real-axis crossing detunings. (a) The normalized self-consistency function for $N = 48$ (points) and its local quadratic expansion (dashed). (b) Negative-minimum depth and its descriptive four-size exponent. (c) Relative error of the quadratic half-gap. The exact eliminated roots have much smaller errors than this local expansion. Depth and curvature are evaluated from the finite-N spectrum.

Across the four tested sizes, Eq. (22) recovers the full-mode residues. Removing BGx_Q while keeping the same roots and denominator gives order-unity or larger relative errors for the high-branch source (SM). Correct rates alone do not fix observable weights.

A physical control isolates the contribution of initial coherences. It compares the fold product state ρ_f with its globally excitation-number-dephased state $\rho_d = \sum_k \Pi_k \rho_f \Pi_k$. The states share their initial $q = 0$ populations, while their difference $\delta\rho_c = \rho_f - \rho_d$ generates a nonzero density transient through its coherences. A source depending only on x_0 would predict zero. Reconstructing this complete transient also requires modes beyond the real pair, whose relative error remains order unity in the late-time window at the $N = 32$ crossing (SM).

VI. FROM SPECTRAL WEIGHTS TO COMPLETE DENSITY TRANSIENTS

A. Preparation selects the contributing modes

Large expansion coefficients and preparation-dependent visibility require a direct dynamical comparison [10, 11, 21]. We compare the two-real-mode sum with full evolution for the preparations in Sec. III, then include the slower oscillatory conjugate pair at $N = 32$. Sparse exponential multiplication supplies the reference without fitting transient amplitudes or rates.

At the crossing, adding the oscillatory pair reduces the high-branch relative L^2 error over $6 \le t \le 12$ from about 25% to 0.5%. The fold-state four-mode error remains about an order of magnitude larger, demonstrating preparation dependence. The SM gives the full error table and displays spectral contributions after removing the common decay envelope; the reported errors use unscaled signals. These comparisons require the oscillatory pair over the tested window, while the real-pair mean decay rate remains of order γ.

B. Fixed-preparation tests across the crossing

To distinguish mode exchange from changes in preparation, we freeze the high-branch difference at $\Delta_*(N)$, denoting it by $\delta\rho^*_{\mathrm{hi}}$. We also use the coherence difference $\delta\rho_c$, with the same single-atom fold seed at every size and detuning. For each fixed source x, the complete density signal is

$$s_N(t, \Delta) = \mathrm{Tr}\left[\frac{N_e}{N} e^{tL(\Delta)} x\right]. \tag{23}$$

We compare $\Delta_* \pm w_N$, where $w_N = 2g_N/|s_a - s_b|$ is the local mixing scale in Eq. (12). The scans cover $N = 24, 32, 40, 48$ and sample $0 \le t \le 12$, without removing any decay envelope.

For the frozen high-branch source at $N = 48$, individual residues change by order 10^{-3} across this interval, whereas the maximum sampled change of the complete density signal is of order 10^{-6} [Fig. 5(c,d)]. Direct propagation of the first two detuning derivatives accurately predicts both endpoint signals from the center, without a fit or spectral truncation. These comparisons support smooth local response despite strong redistribution of spectral weights. The SM gives derivative errors, coherence-source controls, and two-pole and trace-contraction bounds.

Independent atoms produce a larger coherence-source peak than the interacting ensemble at $\Delta_*(32)$ [Fig. 5(a)].

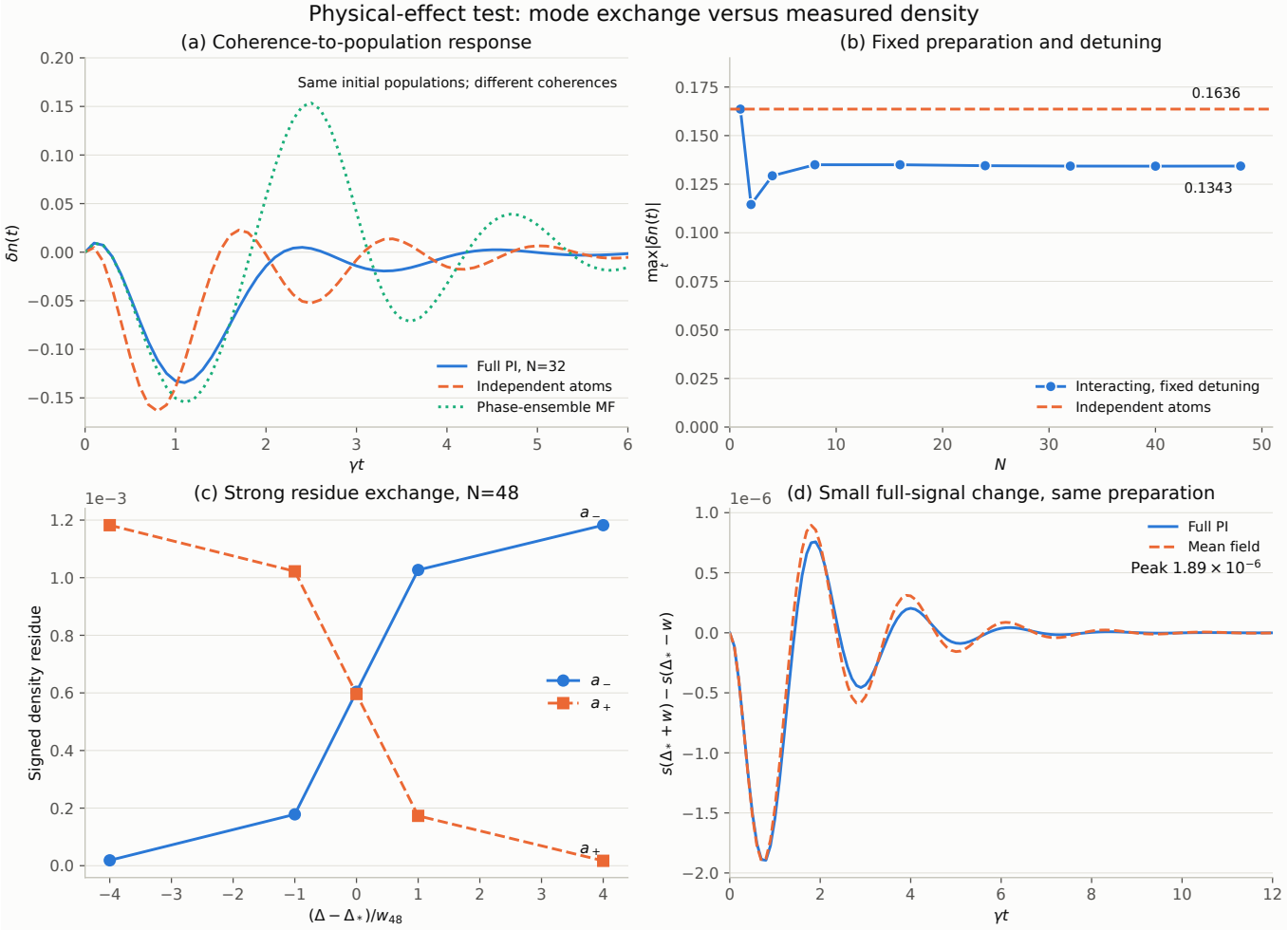


FIG. 5. Fixed-preparation density controls, in raw density units. (a) Fold coherence difference at $N = 32$, $\Delta = \Delta_*(32)$: full permutation-invariant evolution, independent atoms, and a common-phase mean-field ensemble. (b) Largest sampled magnitude at the same detuning and single-atom seed, for $t = 0, 0.1, \ldots, 12$. (c) Signed real-mode residues for the $N = 48$ high-branch difference frozen at its crossing. (d) Difference between the complete signals at $\Delta_* + w_{48}$ and $\Delta_* - w_{48}$ for that fixed preparation. Lines in (b,c) guide the eye; residues in (c) are decomposition weights, whereas (d) compares physical density signals.

At this fixed detuning and single-atom seed, the interacting peak stays near 0.13 over $N = 8$–48 [Fig. 5(b)]. This comparison identifies coherence-to-population conversion as a response already present in independent atoms, with no peak enhancement from the interacting crossing in this control. The global initial trace distance varies across sizes (SM).

The complete coherence-source transient distinguishes the many-body evolution from mean field [Fig. 5(a)]. The mean-field control evolves each common-phase product state separately before averaging, retaining the finite-N interaction factor $(N-1)/N$. Reconstructing the complete transient requires contributions beyond the target pair, whose standalone error for this preparation is already large. Together, these controls characterize spectral redistribution and complete density response for the stated preparations, sizes, and time window.

VII. DISCUSSION AND CONCLUSIONS

The locally damped Rydberg ensemble links finite-rate spectral hybridization to preparation-dependent relaxation through exact coherence elimination. Earlier collective-spin studies established finite-rate doublets and cancelling residues [20, 21]; here their detuning-dependent response is resolved with all local-decay spin blocks retained. Two self-consistent roots determine the rates, while the initial-coherence source and derivative normalization determine their observable weights.

Fixed preparations reveal pronounced residue exchange alongside smooth complete density transients. The independent-atom reference exceeds the interacting coherence-source peak, and the fixed-detuning peak stays nearly size independent over the tested range. The full time dependence nevertheless distinguishes the

many-body evolution from the common-phase mean-field ensemble. These comparisons distinguish spectral hybridization, coherence conversion, and complete relaxation. Identifying the spectral origin of the mean-field discrepancy requires analysis of the remaining mode contributions.

The coupling product gives $g_N^2 = V_{ab}V_{ba}$ exactly, while the local quadratic expansion gives $g_N^2 \simeq 2\epsilon_N/c_N$ over the computed window. An independent asymptotic prediction requires these quantities beyond the exact finite-size pair. Exact steady-state information [12] and the purification-barrier conjecture for the ordinary Liouvillian gap [16] address a different spectral scale. Switching-instanton theory [14, 15] would require boundary conditions matched to this excited pair. The asymptotic exponent and thermodynamic fold correspondence remain open.

For the finite systems studied here, the central result is a quantitative connection between mode character, coherence memory, and physical response. Combining roots, source terms, and direct propagation makes that connection explicit and resolves how sharp spectral redistribution coexists with smooth observable relaxation.

DATA AND NUMERICAL METHODS

The SM specifies the generator, preparations, projection identities, control calculations, and numerical checks. Plotted records, failed numerical attempts and their refinements, and reproduction scripts accompany the source bundle. Independent full-spin checks cover $N = 1$–4, and detuning derivatives are checked against independent small-system time integration. The finite-size calculations use no fitted stochastic noise model.

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